NAJAF SHEHZAD

COMPUTER SCIENCE | AI Specialization
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SUMMARY

Enthusiastic about Artificial Intelligence, Machine Learning, Deep Learning, and Computer Vision, with a strong drive to deepen my expertise in intelligent systems. Skilled in Python, React Native, and SQL, with hands-on experience building mobile applications and working with databases. Committed to continuously strengthening my technical skills to develop modern, scalable, and user-friendly solutions.

TECHNICAL SKILLS :

- Python
- React Native
- SQL
- FastAPI
- Git / GitHub

PROJECTS :

Smart City Overhead Traffic Monitoring and Congestion Analysis Using Computer Vision (FYP) :

Tech Stack: YOLOv8 (Ultralytics), ByteTrack, OpenCV, PaddleOCR, FastAPI, SQLAlchemy and MYSQL, React Native.

- **Problem:** Pakistani city roads are overloaded, and there is no system to monitor and analyse traffic congestion in real time.
- **Detection and Counting:** We collected a custom dataset and fine-tuned YOLOv8 using the Ultralytics library to detect vehicles, classify them, and track each one with built-in ByteTrack so it is counted once as it crosses a virtual line.
- **Number Plate Recognition:** A second Ultralytics YOLO model locates the plate inside each detected vehicle, and PaddleOCR reads the characters after OpenCV preprocessing. Readings from multiple frames are voted on to pick the most reliable plate, and the vehicle colour is also recorded.
- **Speed Tracking and Ticketing:** Cameras are linked by matching the same plate across locations, and average speed is calculated from the camera GPS distance and the elapsed time. Vehicles exceeding the limit are issued overspeed tickets automatically.

EXPERIENCE :

Internship — FlyRank.AI | July–September 2026

Machine Learning :

- Developed an end-to-end content-scoring ML pipeline using 30,000+ records across 32 clients.
- Compared Logistic Regression and Random Forest against a rule-based baseline using client-grouped validation.
- Corrected a 34-point precision inflation from improper data splitting and published reproducible findings in a research paper.

AI Fluency : Profile

- Built a live ML portfolio website with a functional contact-form backend.
- Developed a research agent with self-evaluating search logic, evaluation cases, and guardrails.
- Applied structured prompt engineering and AI-assisted development throughout the project.

TOOLS PROFICIENCY :

VS Code, Claude, ChatGPT, GitHub, Google Collab, Canva, MS Office

CERTIFICATE :

DeepLearning.AI - Supervised Machine Learning: Regression and Classification

FlyRank.AI - AI Fluency

FlyRank.AI - Machine Learning

EDUCATION

Islamabad, Pakistan **01/09/2017 – 16/10/2020**

INTERMEDIATE Punjab Board of Technical Education, Lahore, Pakistan
(CTTI) CONSTRUCTION TECHNOLOGY TRAINING INSTITUTE

Rawalpindi, Pakistan **13/02/2023 – CURRENT**

BACHELORS OF COMPUTER SCIENCE
PMAS-Arid Agriculture University, Rawalpindi

RECOMMENDATION

Director HR, Defence Housing Authority, Gujranwala

Dedicated AI/ML and full-stack developer with strong technical skills and a positive attitude, highly recommended for IT roles in Pakistan and abroad.